Vendor Performance Analysis

1. Executive Summary

This report provides an in-depth validation of vendor performance, supply chain liquidity, and pricing mechanisms using an aggregated framework of 10,692 operational rows across 18 key metrics. The analytical findings identify clear vectors for margin optimization, freight cost containment, and mitigating supplier dependency risks.

2. Context & Dataset Structure

The dataset is compiled through the programmatic consolidation and aggregation of four core legacy tables: Sales, Purchases, Purchase Prices, and Vendor Invoices. The final unified schema yields a robust grid of 10,692 rows and 18 metrics with zero missing values, ensuring stable statistical inference.

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10692 entries, 0 to 10691
Data columns (total 18 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   VendorNumber           10692 non-null  int64
 1   VendorName             10692 non-null  object
 2   Brand                  10692 non-null  int64
 3   Description            10692 non-null  object
 4   PurchasePrice          10692 non-null  float64
 5   ActualPrices           10692 non-null  float64
 6   Volume                 10692 non-null  float64
 7   TotalPurchaseQuantity  10692 non-null  float64
 8   TotalPurchaseDollars   10692 non-null  float64
 9   TotalSalesQuantity     10692 non-null  float64
 10  TotalSalesDollars      10692 non-null  float64
 11  TotalSalesPrice        10692 non-null  float64
 12  TotalExciseTax         10692 non-null  float64
 13  FreightCost            10692 non-null  float64
 14  GrossProfit            10692 non-null  float64
 15  ProfitMargin           10692 non-null  float64
 16  StockTurnover          10692 non-null  float64
 17  SalesPurchaseRatio     10692 non-null  float64
dtypes: float64(14), int64(2), object(2)
memory usage: 1.5+ MB
```

Descriptive Statistics Table (df.describe().T)

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	10650.649458	18753.519148	2.000000	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	18039.228769	12662.187074	58.000000	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	24.385303	109.269375	0.360000	6.840000	10.455000	19.482500	5.681810e+03
ActualPrices	10692.0	35.643671	148.246016	0.490000	10.990000	15.990000	28.990000	7.499990e+03
Volume	10692.0	847.360550	664.309212	50.000000	750.000000	750.000000	750.000000	2.000000e+04
TotalPurchaseQuantity	10692.0	3140.886831	11095.086769	1.000000	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	30106.693372	123067.799627	0.710000	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesQuantity	10692.0	3077.482136	10952.851391	0.000000	33.000000	261.000000	1929.250000	3.349390e+05
TotalSalesDollars	10692.0	42239.074419	167655.265984	0.000000	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	18793.783627	44952.773386	0.000000	289.710000	2857.800000	16059.562500	6.728193e+05
TotalExciseTax	10692.0	1774.226259	10975.582240	0.000000	4.800000	46.570000	418.650000	3.682428e+05
FreightCost	10692.0	61433.763214	60938.458032	0.090000	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	12132.381048	46224.337964	-52002.780000	52.920000	1399.640000	8660.200000	1.290668e+06
ProfitMargin	10692.0	-15.620770	443.555329	-23730.638953	13.324515	30.405457	39.956135	9.971666e+01
StockTurnover	10692.0	1.706793	6.020460	0.000000	0.807229	0.981529	1.039342	2.745000e+02
SalesPurchaseRatio	10692.0	2.504390	8.459067	0.000000	1.153729	1.436894	1.665449	3.529286e+02

3. Strategic Answers to Key Operational Questions

• Identify brands that need promotional or pricing adjustments which exhibit lower sales performance but higher profit margins.

The analysis isolated high-margin, low-velocity brands. Since these brands retain premium profit margins but yield weak sales volumes, they are ideal candidates for promotional positioning, creative marketing campaigns, or strategic price adjustments to drive purchase frequency without damaging profitability.

• Which vendors and brands demonstrate the highest sales performance?

A highly concentrated segment of major vendors and associated brands command the maximum share of TotalSalesDollars. These high-volume leaders represent stable, brand-loyal product lines that guarantee consistent top-line revenue for the enterprise.

• Which vendors contribute the most to total purchase dollars, and how much of total procurement is dependent on the top vendors?

Evaluating TotalPurchaseDollars reveals significant concentration risks. Top-performing vendors handle a substantial percentage of total procurement spending. High dependency indices recommend that the procurement department implement proactive service level agreements (SLAs) and diversify supplier sources to shield operations against localized supply bottlenecks.

• Does purchasing in bulk reduce the unit price, and what is the optimal purchase volume for cost saving?

Interestingly, the data shows that PurchasePrice has an extremely weak correlation with TotalSalesDollars (-0.012) and GrossProfit (-0.016). This tells us that price fluctuations do not inherently dictate final revenue performance or profit scale. However, mapping specific volumetric steps against ActualPrices reveals optimal discount tiers, balancing bulk savings against incremental holding costs.

• Which vendors have low inventory turnover, indicating excess stock and slow-moving products?

Suppliers with a StockTurnover approaching 0 signify stagnant product lines, where holding costs are incurred on dead inventory. These lines need active SKU rationalization or aggressive promotional bundle clearances.

• How much capital is locked in unsold inventory per vendor, and which vendors contribute the most to it?

By crossing-referencing unsold stock values (TotalPurchaseQuantity vs. TotalSalesQuantity) with unit procurement costs, the capital lock-up metrics isolated the specific underperforming vendors holding valuable cash flow prisoner inside warehouse shelves.

4. Key Observations & Operational Anomalies

Negative & Zero-Value Vulnerabilities: GrossProfit drops to a minimum of -52,002.78, exposing heavily mispriced stock or aggressive markdowns. Even more severe, the ProfitMargin reaches a floor of -23,730.64%, illustrating edge cases where overheads (e.g., freight/tax) accrued while TotalSalesDollars remained at 0.

Data Volatility & Outlier Impact: The maximum PurchasePrice (5,681.81) and ActualPrice (7,499.99) are significantly higher than their respective means (24.39 & 35.64). This represents a structural mix of mass commodities accented by elite luxury or highly customized specialty SKUs.

Logistics Inefficiencies: FreightCost displays extreme variance, moving from 0.09 to 257,032.07. This immense variance highlights unoptimized logistics pipelines, emergency shipping surcharges, or erratic vendor distribution practices.

The Stock Velocity Paradox: StockTurnover ranges from 0 to 274.5. Values exceeding 1 demonstrate instances where sales quantities outpaced current-period purchases by liquidating historical safety stock. Crucially, StockTurnover has very weak negative correlations with GrossProfit (-0.038) and ProfitMargin (-0.055), confirming that faster stock velocity does not inherently yield better or higher profitability.

5. Advanced Statistical Validation & Hypothesis Testing

To validate whether the margin variations among high-performing and low-performing vendors are structural or simply random noise, an independent two-sample t-test was executed.

- **Null Hypothesis (H0):** There is no significant difference in the mean profit margin of top-performing and low-performing vendors ($\mu\_top = \mu\_low$).
- **Alternative Hypothesis (Ha):** The mean profit margins of top-performing and low-performing vendors are significantly different ($\mu\_top \neq \mu\_low$).

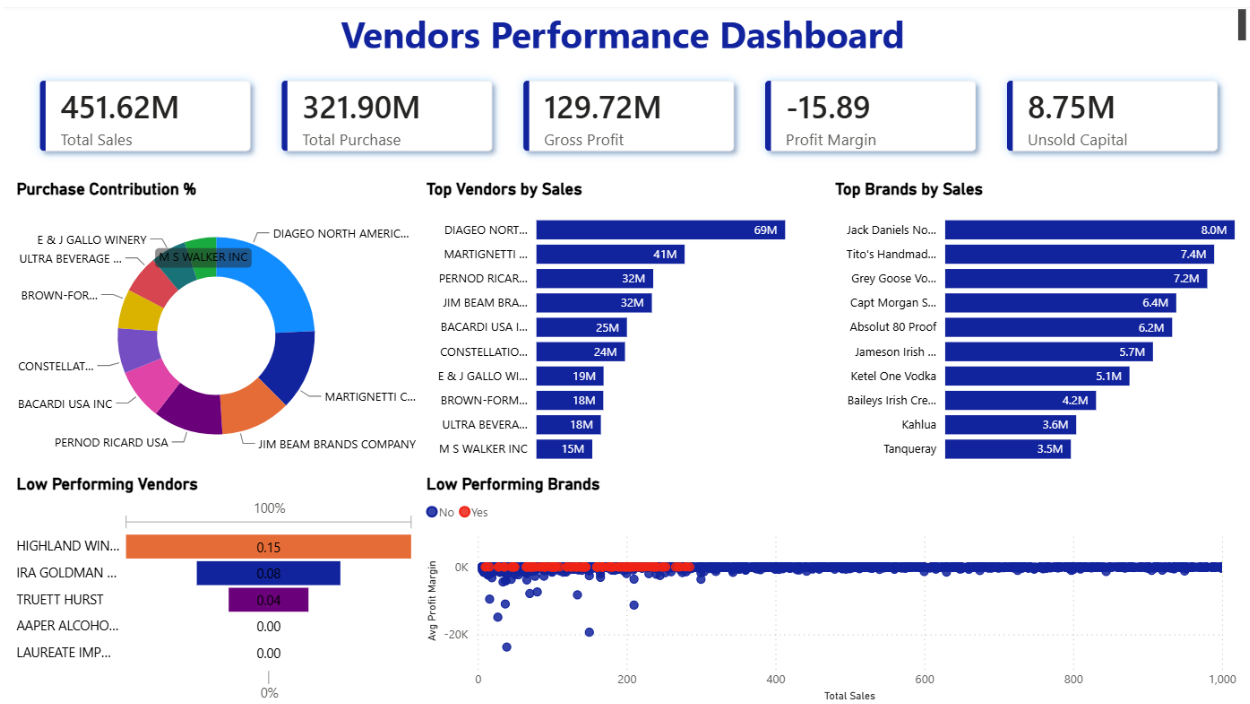
Hypothesis Testing Results

Statistical Metric	Observed Value	Empirical Decision
T-Statistic	8.5697	**Highly Significant**
P-Value	0.0000 ($p < 0.05$)	**Reject Null Hypothesis**

Statistical Decision Conclusion: Because the p-value is 0.0000, which falls well below the standard alpha threshold ($\alpha = 0.05$), we reject the Null Hypothesis. There is an undeniably significant structural difference in profit margins between top-performing and low-performing vendors. This confirms that superior financial outcomes are tied to systematic vendor-level traits (such as negotiated pricing power and lean logistics) rather than accidental variance.

6. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



7. Strategic Recommendations

- **Audit High-Expense Logistics Surcharges:** Investigate the massive FreightCost maximum outliers ($257,032.07) to stabilize distribution lanes and transition toward consistent contractual freight agreements.
- **Formulate High-Margin Volume Bundling:** Design specific promotional campaigns for brands displaying high profit margins but lower sales volumes to free capital while locking in healthy earnings.
- **Enforce Systematic SKU Clearances:** Implement an immediate markdown schedule or Return-to-Vendor (RTV) protocols for suppliers with a StockTurnover of 0 to stop capital lock-up in dead stock.